SESSION 4

FINANCIAL RATIO ANALYSIS

SCALER SCHOOL OF TECHNOLOGY

Learning Outcomes

Session 4: Financial Ratio Analysis

By the end of the session, learners will be able to:

- **Explain** the purpose and importance of financial ratio analysis.
- **Describe** key profitability, return, efficiency, working capital, liquidity, and leverage ratios.
- **Interpret** these ratios to assess a company's performance, financial health, efficiency, returns, and risk.
- **Apply** financial ratios to support business evaluation and decision-making.

Financial Ratios: Introduction

Financial ratios are critical metrics used to **evaluate** a **company's financial health** and **performance.**

✔ It is derived from a firm's financial statements.

✔ These ratios facilitate comparisons both within the company over time and against industry peers.

Objectives of financial ratios:

- ❑ To **assess profitability** – whether the business generates **sufficient profits** for its owners and other stakeholders
- ❑ To **evaluate liquidity** – whether the company can meet its **short-term obligations.**
- ❑ To **measure operational efficiency** – whether the company uses its assets efficiently
- ❑ To **assess solvency** and **financial risk** – whether the company can meet its **long-term obligations**
- ❑ To **evaluate returns on capital employed** – provided by **owners and lenders**
- ❑ To facilitate **performance comparisons** – across **time, competitors, industry & budgeted figures**
- ❑ To support **managerial decision-making**

Financial Ratios: Types

Financial ratios can be grouped into five major categories:

- **Liquidity Ratios –** Can the business meet its **short-term obligations?**
- **Efficiency Ratios –** How **efficiently** is the business using its **assets** and working capital?
- **Solvency/Leverage Ratios –** Can the business meet its **long-term obligations**, and how much financial risk does it carry?
- **Profitability & Return Ratios –** How effectively is the **business generating profit** and returns?
- **Market/Investor Ratios –** How is the **business performing** from the perspective of shareholders and the stock market?

Profitability & Return Ratios

Profitability & Return Ratios: They measure a company's ability to generate **profit from its sales, assets, capital employed and shareholders' funds.**

"How effectively is the company converting its revenue and invested resources into profit and returns?"

A company may have:

- Strong liquidity but poor profitability
- High sales but low profit margins
- High profits but poor returns on capital
- High returns because of excessive financial leverage

Therefore, profitability analysis should not focus on **profit alone.** It should examine **profit relative to sales, assets and capital invested.**

They can be divided into two broad groups:

- **Profit Margin Ratios:** These measure profit relative to revenue.
- **Return Ratios:** These measure profit relative to capital, assets or equity.

Profitability & Return Ratios

Profitability Ratios: They focus primarily on the company's ability to generate **profit from revenue.**

Types of Profitability Ratios:

- **Gross Profit Margin (GPM):** It measures the **percentage of revenue** remaining after **deducting the cost of goods sold/cost of sales.**
  - ✔ It evaluates the **profitability of the company's** basic production or purchasing activity **before operating expenses, finance costs and taxes.**

Gross Profit Margin Ratio is calculated as follows:

$$\text{Gross Profit Margin} = \frac{\text{Gross Profit}}{\text{Revenue}} \times 100$$

Where,

$$\text{Gross Profit} = \text{Revenue} - \text{Cost of Sales}$$

Therefore, Gross Profit Margin =

$$\text{Gross Profit Margin} = \frac{\text{Revenue} - \text{Cost of Sales}}{\text{Revenue}} \times 100$$

Profitability & Return Ratios

Types of Profitability Ratios:

- **Operating Profit Margin (OPM):** Also called **EBIT (Earnings Before Interest & Tax) after operating expenses and depreciation/amortisation but before interest and tax.**

Operating Profit Margin is calculated as follows:

$$Operating\ Profit\ Margin = \frac{EBIT}{Revenue} \times 100$$

- **Net Profit Margin (NPM):** It measures the percentage of revenue that remains as profit after considering the company's relevant expenses, including operating expenses, finance costs and taxation.

Net Profit Margin is calculated as follows:

$$Net\ Profit\ Margin = \frac{Profit\ After\ Tax}{Revenue} \times 100$$

Profitability & Return Ratios

Return Ratios: They measure the returns generated from the resources invested in the business.

Major Types of Return Ratios:

- **Return on Capital Employed (ROCE):** It measures how **efficiently** a company generates operating profit from the **long-term capital employed** in the business.

ROCE is calculated as follows:

$$ROCE = \frac{EBIT}{Capital\ Employed} \times 100$$

Where,

$$Capital\ Employed = Total\ Assets - Current\ Liabilities$$

Alternatively, Capital Employed can **also** be calculated as follows:

$$Capital\ Employed = Equity + Long\text{-}term\ Debt$$

Profitability & Return Ratios

Major Types of Return Ratios:

- **Return on Assets (ROA):** It measures how **efficiently** a company uses its **asset base** to generate **profits.**

ROA is calculated as follows:

$$ROA = \frac{Profit\ After\ Tax}{Average\ Total\ Assets} \times 100$$

Where,

$$Average\ Assets = \frac{Opening\ Assets + Closing\ Assets}{2}$$

- **Return on Equity (ROE):** It measures the **profit** generated from the **equity capital** invested by shareholders.

ROE is calculated as follows:

$$ROE = \frac{Profit\ After\ Tax}{Average\ Shareholders'\ Equity} \times 100$$

Where: **Shareholders' Equity = Share Capital + Retained Earnings (undistributed profits belonging to shareholders)**

Profitability & Return Ratios

Major Types of Return Ratios:

- ❑ **Return on Investment (ROI):** It is a broader measure used to evaluate the return generated from a particular investment.
- ✔ Unlike ROE or ROCE, ROI is not always calculated using one universally accepted accounting formula.

ROI is calculated as follows:

$$ROI = \frac{\text{Gain from Investment} - \text{Investment Cost}}{\text{Investment Cost}} \times 100$$

Profitability & Return Ratios

Market/Investor Ratios: These ratios are primarily used to evaluate a company's performance from the perspective of **equity shareholders and the stock market.**

"How does the stock market value the company's earnings, assets and future growth prospects?"

OR

"What return or value is the company generating for its shareholders?"

They often incorporate information from the stock market:

- Market price per share
- Market capitalisation **(Market Cap)**
- Earnings per share **(EPS)**
- Dividends per share **(DPS)**
- Book value per share **(BV)**

It helps investors of listed companies to assess:

- The market's valuation of the company – whether the share is undervalued or overvalued
- Earnings generated per share
- Dividend & growth expectations

Profitability & Return Ratios

Types of Market Ratios:

- **Earnings Per Share (EPS):** It measures the amount of **profit attributable** to **ordinary shareholders** for each ordinary share.

EPS is calculated as follows:

$$EPS = \frac{Profit\ After\ Tax - Preference\ Dividend}{Number\ of\ Ordinary\ Shares}$$

OR

$$EPS = \frac{Profit\ Attributable\ to\ Ordinary\ Shareholders}{Number\ of\ Ordinary\ Shares}$$

- **Price-Earnings Ratio (P/E):** It measures how much **investors are willing to pay for each unit of a company's current earnings.**

P/E Ratio is calculated as follows:

$$P/ERatio = \frac{\text{Market Price Per Share}}{\text{EPS}}$$

Profitability & Return Ratios

Types of Market Ratios:

- **Dividend Per Share (EPS):** It measures the **dividend distributed to shareholders** for each ordinary share.

DPS is calculated as follows:

$$DPS = \frac{\text{Total Ordinary Dividends}}{\text{Number of Ordinary Shares}}$$

- **Dividend Payout Ratio:** It measures the **proportion of earnings distributed** to shareholders as dividends.

Dividend Payout Ratio is calculated as follows:

$$Dividend\ Payout\ Ratio = \frac{Dividend\ Per\ Share}{EPS} \times 100$$

Alternatively, Dividend Payout can **also** be calculated as follows:

$$Dividend\ Payout\ Ratio = \frac{Total\ Dividends}{Profit\ Attributable\ to\ Ordinary\ Shareholders} \times 100$$

Profitability & Return Ratios

Types of Market Ratios:

- **Dividend Yield:** It measures the **annual dividend income from a share as a percentage of its current market price.**

Dividend yield is calculated as follows:

$$Dividend\ Yield = \frac{Dividend\ Per\ Share}{Market\ Price\ Per\ Share} \times 100$$

- **Market Capitalisation:** Although **not technically a ratio, market capitalisation** is a fundamental market-value measure and is often analysed alongside investor ratios.

It is represented by the formula:

$$Market\ Capitalisation = Market\ Price\ Per\ Share \times Number\ of\ Shares\ Outstanding$$

- **Enterprise Value (EV):** It tells us **how much the entire business is worth,** considering both the money invested by shareholders and the money borrowed by the company, while **subtracting the company's cash.**

✔ It is widely used in **business valuation, mergers and acquisitions, and investment analysis.**

It is represented by the formula:

$$Market\ Capitalisation = Market\ Price\ Per\ Share \times Number\ of\ Shares\ Outstanding$$

Thank You!